Álvaro Liébana Carrascosa

Valencia, Spain
+34 638 933 718 • alvaro@tricubo.tech
alvaroliebana.com, LinkedIn, github.com/leyenda95

Full-stack product engineer with hands-on experience at the blockchain layer (Ethereum, Algorand, Cardano and Midnight). I build interfaces, backend logic and wallet-facing flows using end-to-end encryption and zero-knowledge proofs. I also review technical proposals for Project Catalyst, assessing feasibility, budget and impact. I turn product needs into features a team can build, and I can explain them clearly to both technical and non-technical audiences. My strength is bridging engineering and product: I start from a real user problem, decide what's worth building, and take it through to a working product.

Core Competencies

- **Blockchain Product and Architecture**: product vision and roadmap; architecture across Ethereum/EVM, Algorand, Cardano and Midnight; choosing the network and design that fit the use case.
- **Smart Contracts**: Solidity on Ethereum; contracts on Cardano and Algorand through their TypeScript variants; Compact on Midnight.
- **Privacy and Proof Systems**: zero-knowledge proofs, selective disclosure and end-to-end encryption; privacy layer written in Compact (Midnight.js).
- **Requirements to Delivery**: turning business and technical needs into clear requirements and driving them to delivery through engineering teams and external parties, including third-party integrations.
- **AI in the Product and Workflow**: incorporation of AI agents into the development workflow, with tests and CI quality gates (coverage thresholds, mutation testing).
- **Tools**: technical writing; TypeScript, Python, Docker, GitHub, Notion, ClickUp.

Professional Experience

- **Aegis** — **Remote**
  Co-founder, Blockchain Lead — *May 2026 - Present*
  Privacy-preserving retail market intelligence on Midnight.
  - Design and implementation of the privacy layer in Compact: each purchase proven with zero-knowledge proofs, and AI agents turning the aggregated signals into market insights and recommendations for each store.
  - Product approach: small retailers compete in the dark against large chains, and shoppers do not want to be tracked. Aegis gives every store collective market intelligence without sacrificing anyone's privacy.

- **Tricubo Technologies** — **Hybrid**
  Blockchain Architect — *2023 - 2026*
  Blockchain products for retail: private receipts and rewards.
  - As blockchain architect, designed the receipt flow: the store issues each receipt at checkout, and it reaches the customer's mobile encrypted end to end on Algorand, readable only by the customer and the store.
  - Integration with Shopify to extend the same flow to e-commerce.
  - Product vision, roadmap, business development and partnerships, taking the product from idea to delivery.

- **Project Catalyst, Cardano** — **Remote**
  Funding Proposal Reviewer — *2026*
  - Assessment of Cardano ecosystem funding proposals against technical feasibility, budget and impact criteria.
  - Written reasoning for each assessment, so that proposers and voters can act on it.

- **Zunibal** **Spain**
  Solutions Architect and Software Engineer *Aug 2021 - Oct 2024*
  Acoustic systems for biomass estimation from satellite buoys.
  - Design of detection and signal processing algorithms, implemented on different computing architectures.
  - Build, maintenance and repair of marine hardware prototypes: sensors, sensor networks, underwater cameras and instrumentation.
- **Spanish State Research Agency (AEI)** **Valencia, Spain**
  Systems and Signal Processing Developer *Feb 2021 - Jun 2021*
  Acoustic detection with neutrino telescopes.
  - Development and operation of underwater acoustic detection systems: lab and open-sea calibration of scientific echosounders, signal processing algorithms and hydroacoustic equipment.
- **Spanish State Research Agency (AEI)** **Valencia, Spain**
  IoT Infrastructure Engineer *Apr 2020 - Dec 2020*
  Software-defined cognitive network for IoT.
  - Testbed setup, multi-vendor network configuration, network measurement and analysis, and technical documentation.

Education

- **Universitat Politècnica de València** **Valencia, Spain**
  Master's in Communication Technologies, Systems and Networks *Sep 2019 - Nov 2020*
- **Universitat Politècnica de València** **Valencia, Spain**
  Bachelor's in Telecommunications Engineering *Sep 2015 - Jun 2019*

Certifications

- **Midnight Scholar**, Midnight Academy (2025).
- **Cardano Blockchain Certified Associate**, Cardano Academy (2025).

Languages

- Spanish (native), Catalan (native), English (professional working proficiency).

References

- Available upon request.